

WEEK 4 – ASSIGNMENT

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DIT University

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

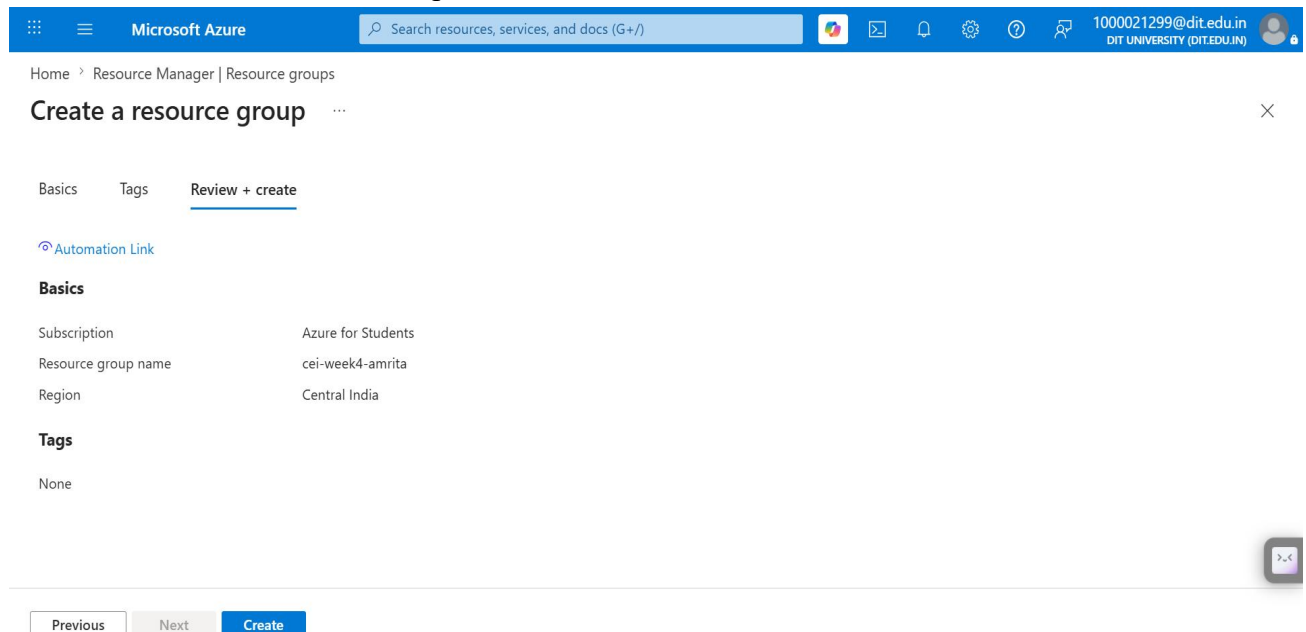
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



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Microsoft Azure

Search resources, services, and docs (G+/I)

1000021299@dit.edu.in
DIT UNIVERSITY (DIT.EDU.IN)

Home > Resource Manager

Resource Manager | Resource groups

How to manage changes with deployment tools? Export resource groups using Bicep or Terraform +1

Search

Create Manage view Refresh Export to CSV Open query Assign tags Group by none

Filter for any field...

Subscription equals all Location equals all Add filter

Name ↑	Subscription	Location
cei-week4-amrita	Azure for Students	Central India

Showing 1 - 1 of 1. Display count: auto

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

Microsoft Azure

Search resources, services, and docs (G+/I)

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Home > ceistorage01_1783878409415 | Overview

ceistorage01 Storage account

Help me reduce the costs of my storage account How can I make my storage account more resilient? Check data resiliency for storage accounts

Search

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

Activity log Tags Diagnose and solve problems Access Control (IAM) Data migration Events Storage browser Storage Mover Partner solutions Resource visualizer Data storage

Essentials

Resource group (move) cei-week4-amrita

Location centralindia

Subscription (move) Azure for Students

Subscription ID f64cbfc8-f763-4e7c-9d6a-b8c9f19cae95

Disk state Available

Tags (edit) Add tags

Performance Standard

Replication Locally redundant storage (LRS)

Account kind StorageV2 (general purpose v2)

Provisioning state Succeeded

Created 12/07/2026, 23:17:31

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

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Microsoft Azure

Search resources, services, and docs (G+)

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Home > ceistorage01_1783878409415 | Overview > ceistorage01

ceistorage01 | Containers

Storage account

Search

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Data management

Settings

Add or remove favorites by pressing Ctrl+Shift+F

Actions: Add container, Upload, Refresh, Delete, Change access level, Restore containers, Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	12/07/2026, 23:18:01	Private	Available	-
☐	processed-data	12/07/2026, 23:26:02	Private	Available	-
☐	source-data	12/07/2026, 23:25:25	Private	Available	-

Microsoft Azure

Search resources, services, and docs (G+)

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Home > ceistorage01_1783878409415 | Overview > ceistorage01 | Containers

source-data

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Actions: Add Directory, Upload, Change access level, Refresh, Delete, Copy, Paste, Rename, Acquire lease

source-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample - S...	12/07/2026, 23:28:51	Hot (Inferred)	Block blob	2.18 MiB	Available

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Task 3: ADF Basics

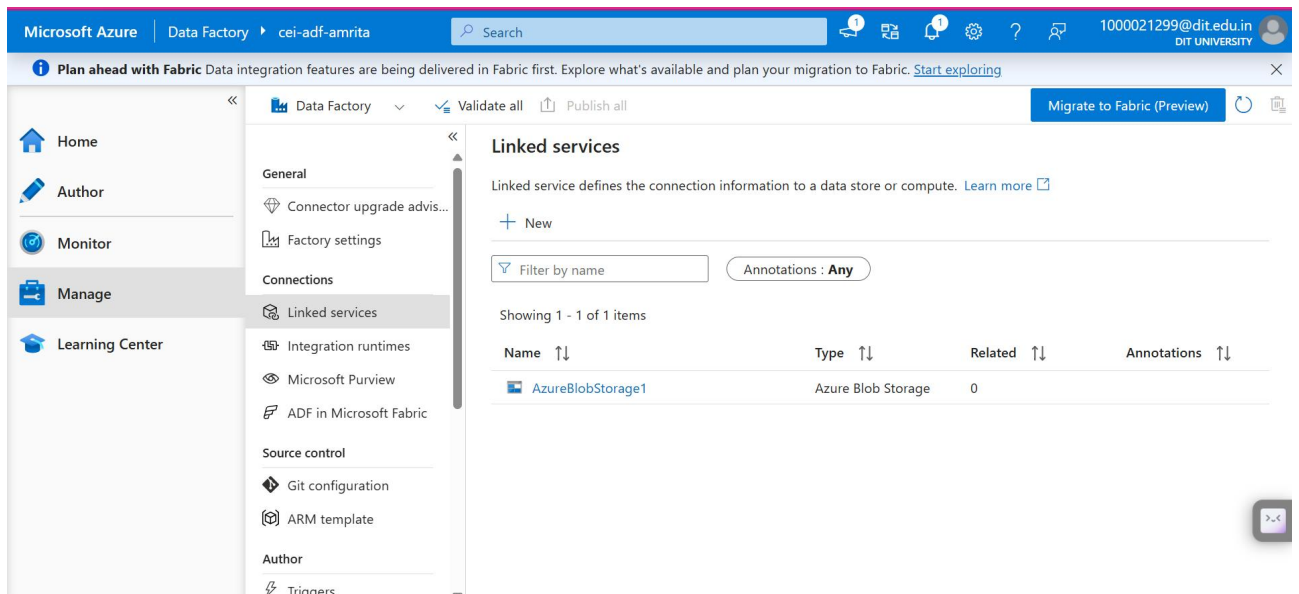
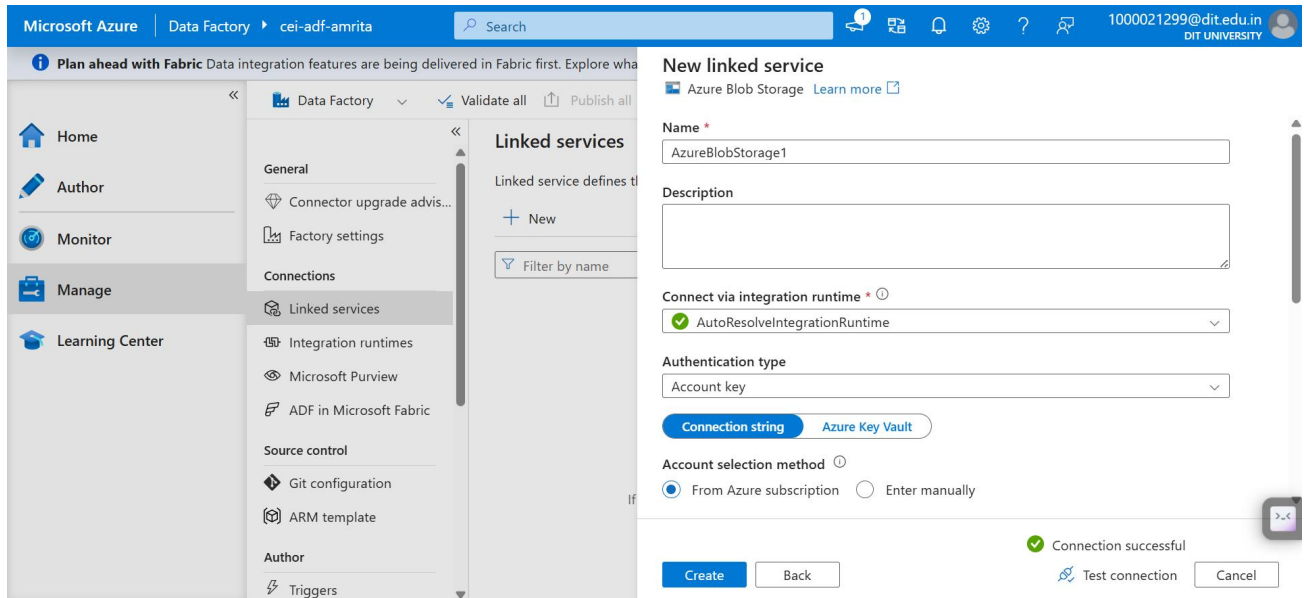
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

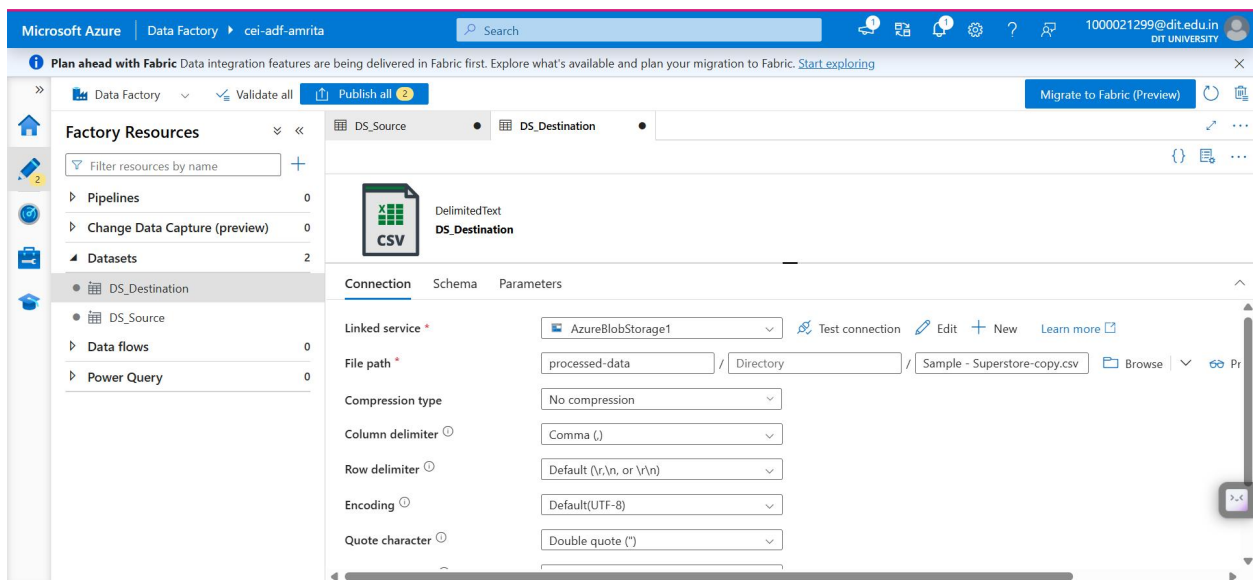
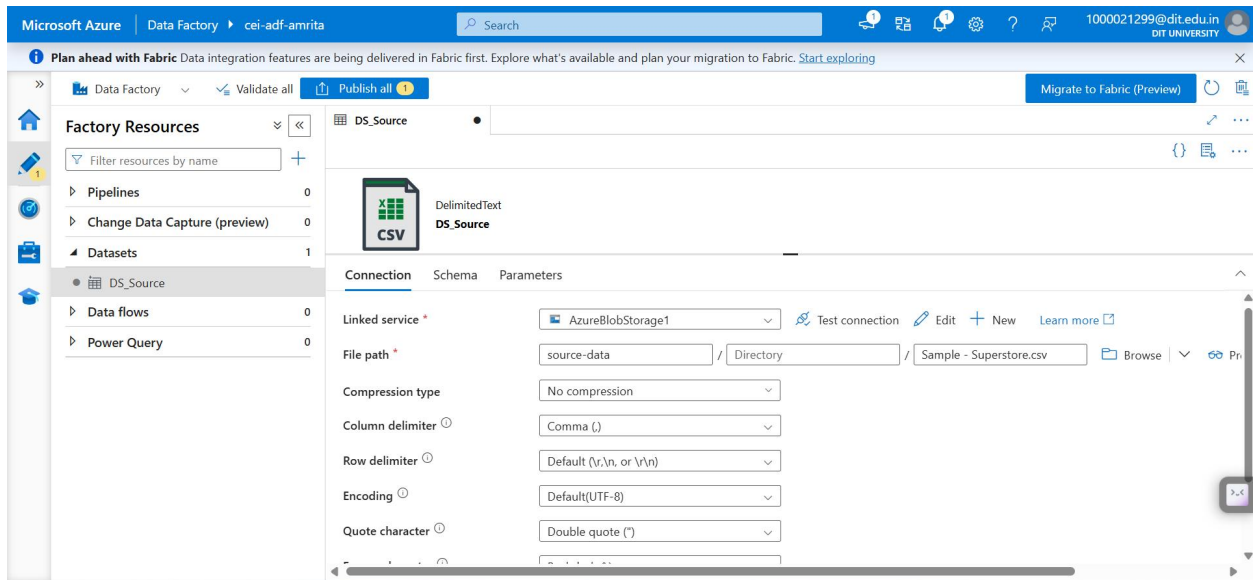
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- Screenshot of Linked Service



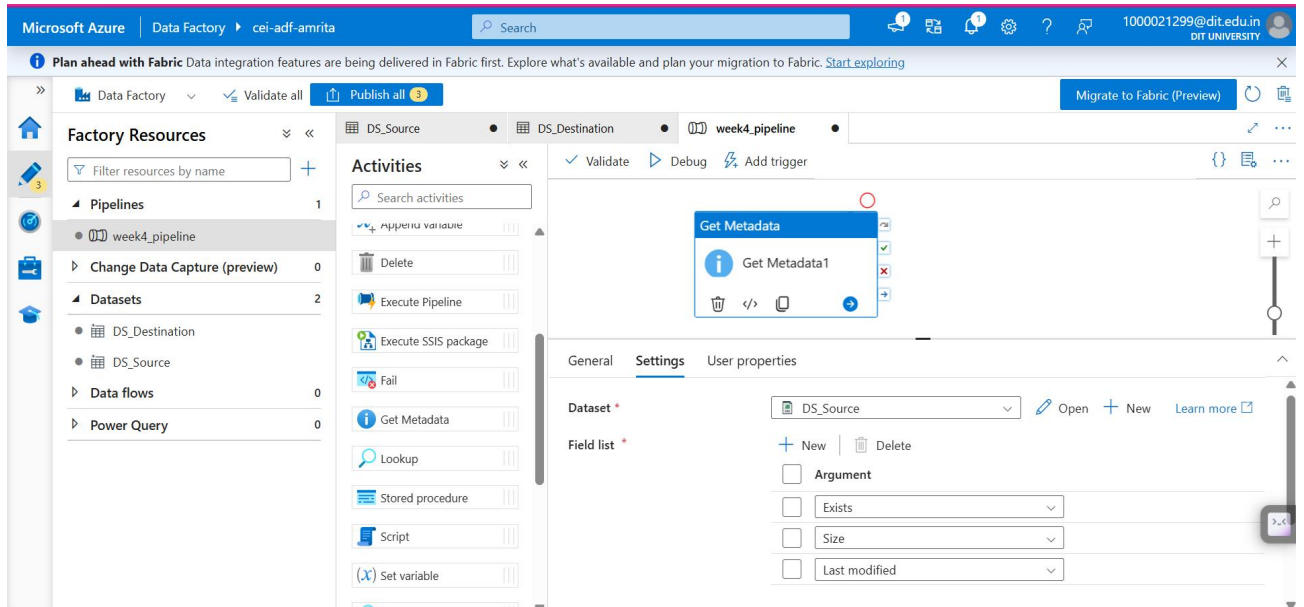
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- Screenshot of dataset



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- Screenshot of Get Metadata activity



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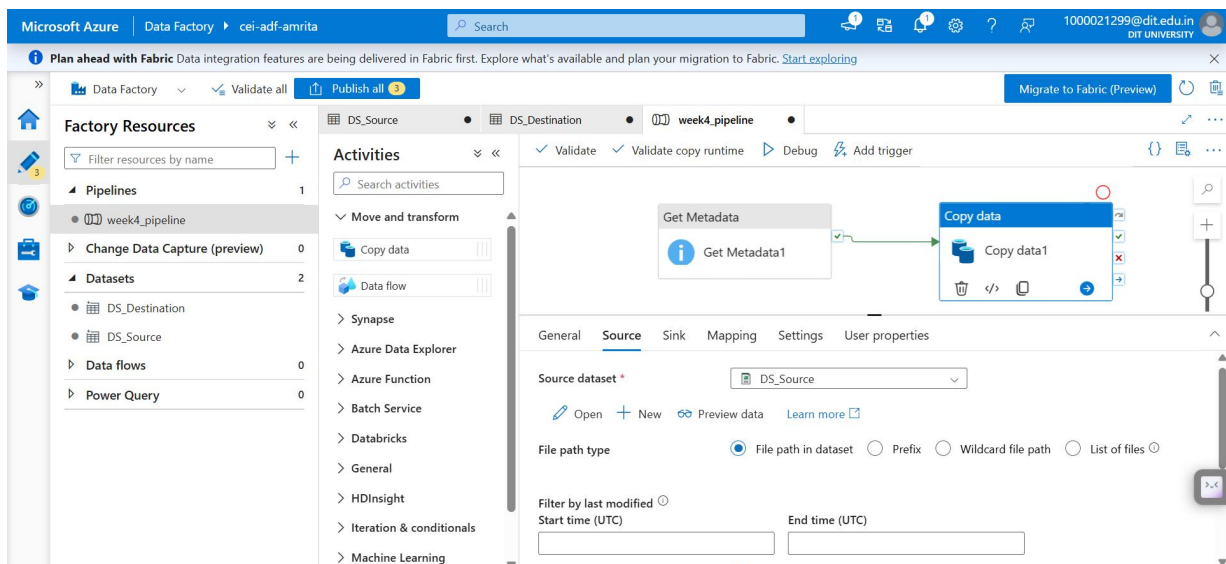
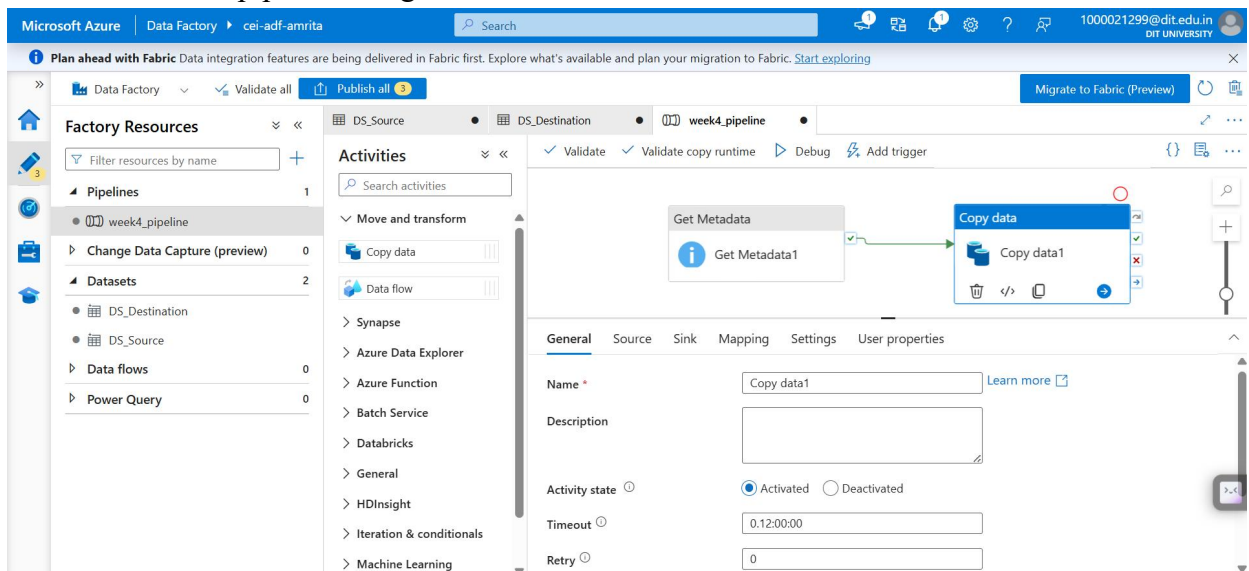
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



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The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar shows 'Microsoft Azure' and 'Data Factory' with the user 'cei-adf-amrita'. A search bar and a 'Migrate to Fabric (Preview)' button are also visible. The main interface is divided into three sections: 'Factory Resources', 'Activities', and a central canvas.

Factory Resources: This section on the left lists the resources available in the factory. It includes a search bar and a list of resources categorized by type: Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'week4_pipeline' is listed under Pipelines. The 'DS_Source' and 'DS_Destination' datasets are listed under Datasets.

Activities: This section in the middle lists the activities available for the pipeline. It includes a search bar and a list of activities: Copy data, Data flow, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, and Machine Learning.

Canvas: The central canvas shows the pipeline 'week4_pipeline' being configured. It contains two activities: 'Get Metadata' (labeled 'Get Metadata1') and 'Copy data' (labeled 'Copy data1'). The 'Copy data' activity is selected, and its properties are displayed in the right-hand pane. The 'Sink dataset' is set to 'DS_Destination'. The 'Copy behavior' is set to 'Select...'. The 'Max concurrent connections' and 'Block size (MB)' are both set to '1'. The 'Metadata' section is expanded, showing 'Quote all text' checked.

The screenshot displays the Microsoft Azure Data Factory console with the 'Factory validation output' dialog open. The dialog shows a green checkmark icon and the text 'Your factory has been validated. No errors were found.' A 'Close' button is located at the bottom right of the dialog.

The background interface is the same as the previous screenshot, showing the 'week4_pipeline' configuration. The 'Factory Resources' section on the left lists the resources available in the factory. The 'Activities' section in the middle lists the activities available for the pipeline. The central canvas shows the pipeline 'week4_pipeline' being configured. The 'Copy data' activity is selected, and its properties are displayed in the right-hand pane. The 'Sink dataset' is set to 'DS_Destination'. The 'Copy behavior' is set to 'Select...'. The 'Max concurrent connections' and 'Block size (MB)' are both set to '1'. The 'Metadata' section is expanded, showing 'Quote all text' checked.

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Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

The screenshot displays the Azure Data Factory console for a pipeline named 'week4_pipeline'. The pipeline is shown in a visual editor with two activities: 'Get Metadata1' and 'Copy data1', both marked with green checkmarks indicating success. Below the visual editor, the 'Output' tab shows the pipeline run details. The pipeline run ID is '65883d19-6c41-4fd8-8535-3ea0b02acb69' and the status is 'Succeeded'. A table lists the activities and their execution details.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/13/2026, 12:08:10 AM	16s	AutoResolveIntegrationRuntime (Central India)
Get Metadata1	✓ Succeeded	Get Metadata	7/13/2026, 12:07:57 AM	12s	AutoResolveIntegrationRuntime (Central India)

The screenshot shows the 'Pipeline runs' section in the Azure Data Factory console. The 'List' view is selected, showing a table of pipeline runs. The table has columns for Pipeline name, Run start, Run end, Duration, Status, Triggered by, Run ID, and Parameters. A single run for 'week4_pipeline' is listed with a status of 'Succeeded'.

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
week4_pipeline	7/13/2026, 12:07:50	7/13/2026, 12:08:27	37s	✓ Succeeded	Manual trigger	65883d19-6c41-4fd8-8...	

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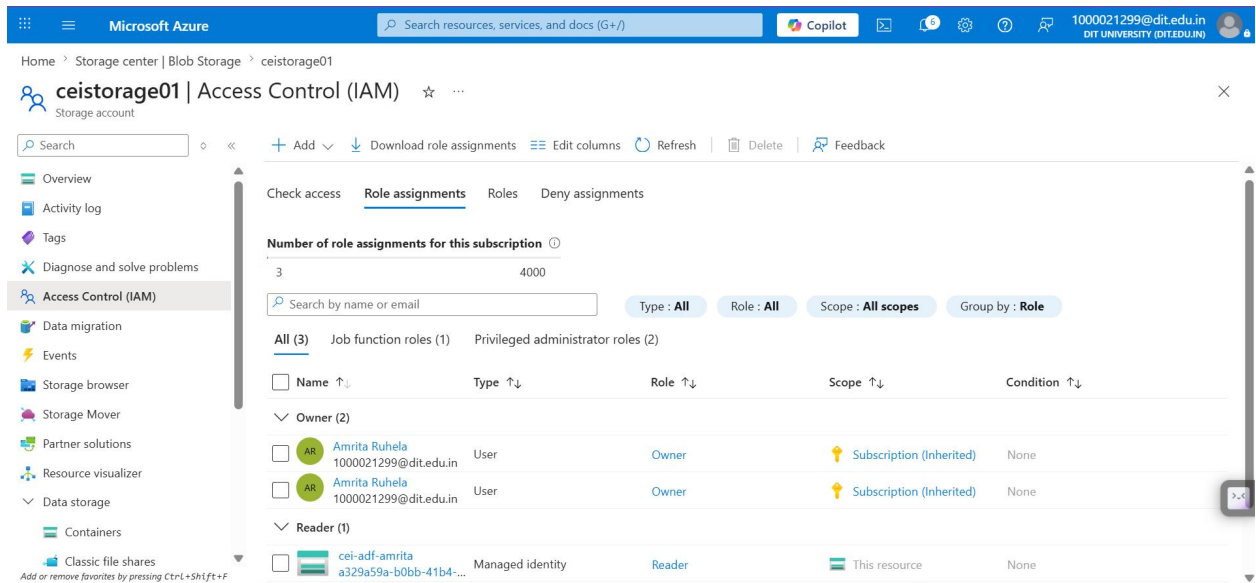
Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



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Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully

The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows 'Microsoft Azure' and 'Data Factory' with the user 'cei-adf-amrita'. A search bar and various icons are present. A notification banner at the top mentions 'Plan ahead with Fabric'. The main section is titled 'Pipeline runs' and includes tabs for 'Triggered' and 'Debug'. Below these are filters for 'Chennai, Kolkata, Mu...' and 'Last 24 hours', along with buttons for 'Pipeline name: All', 'Status: All', and 'Add filter'. A table shows one pipeline run: 'week4_pipeline' with a status of 'Succeeded'. The table columns are Pipeline name, Run start, Run end, Duration, Status, Triggered by, Run ID, and Parameters. The run started on 7/13/2026 at 12:07:50 and ended at 12:08:27, lasting 37 seconds. The status is 'Succeeded' with a green checkmark. The triggered by is 'Manual trigger' and the run ID is '65883d19-6c41-4fd8-8...'. The interface also includes a sidebar with navigation icons and a bottom right corner with a small icon.

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
week4_pipeline	7/13/2026, 12:07:50	7/13/2026, 12:08:27	37s	Succeeded	Manual trigger	65883d19-6c41-4fd8-8...	

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- Data copied to destination

The screenshot shows the Microsoft Azure Storage Explorer interface. The breadcrumb path is Home > Storage center | Blob Storage > ceistorage01 | Containers. The selected container is 'processed-data'. The authentication method is 'Access key'. A search bar is present with the text 'Search blobs by prefix (case-sensitive)'. Below the search bar, it says 'Showing all 1 items'. A table lists the contents of the container:

	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	Sample - Superstore-copy.csv	13/07/2026, 00:08:23	Hot (Inferred)	Block blob	2.18 MiB	Available	...

At the bottom, there is a small text note: 'Add or remove favorites by pressing Ctrl+Shift+F'.

- Metadata validated

The screenshot shows the Microsoft Azure Data Factory interface. The breadcrumb path is Data Factory > cei-adf-amrita. The selected pipeline is 'week4_pipeline'. The pipeline is in the 'Validate' state. The activities are 'Get Metadata1' and 'Copy data1'. The pipeline status is 'Succeeded'. The output table shows the following details:

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/13/2026, 12:08:10 AM	16s	AutoResolveIntegrationRuntime (Central India)
Get Metadata1	✓ Succeeded	Get Metadata	7/13/2026, 12:07:57 AM	12s	AutoResolveIntegrationRuntime (Central India)

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Summary:

In this project, an Azure Resource Group, Storage Account, Blob Storage containers, and Azure Data Factory were successfully created and configured. Linked Services and Datasets were established to connect Azure Data Factory with Azure Blob Storage. A pipeline was developed using the Get Metadata activity to validate the source file's existence, size, and last modified timestamp, followed by the Copy Data activity to transfer the CSV file from the source container to the destination container. Appropriate IAM roles (Reader and Contributor) were assigned to ensure secure access between Azure Data Factory and Azure Storage. The pipeline was validated, published, executed successfully, and the output was verified by confirming the copied file in the destination container along with successful metadata validation.

Outcome:

- Successfully created and configured Azure cloud resources.
- Connected Azure Blob Storage with Azure Data Factory using Linked Services.
- Validated source file metadata before data movement.
- Successfully copied the CSV file to the destination container.
- Verified pipeline execution and destination output.
- Demonstrated an end-to-end Azure Data Factory ETL workflow using Azure cloud services.